ACCURATE-Churn: An Actionable, Calibrated, and Uncertainty-Aware Ensemble Framework for Telecom Customer Churn Prediction and Retention Decision Support

Dataset Description

The experiment used the IBM Telco Customer Churn dataset from Kaggle.

Dataset characteristics

- Total rows: **7043**
- Total columns: **21**
- Target variable: **Churn**
- Churn distribution:
  - No: **5174**
  - Yes: **1869**
- Churn rate: **26.54%**

Main feature groups

The dataset contains:

- demographic information
- account information
- subscribed service information
- billing/payment features
- target churn outcome

Important initial findings

- customerID was unique for all rows and treated only as an identifier.
- TotalCharges was stored as object/string and contained **11 blank values**.
- After cleaning, TotalCharges became numeric.
- The 11 blank TotalCharges rows all had:
  - tenure = 0
  - Churn = No

This indicates that these rows represent very new customers with no accumulated billing history.

Research Objective

The main objective of this experiment was to build a **publishable, practical, and novel churn prediction framework** with the following properties:

- strong predictive performance
- calibrated probabilities
- uncertainty-aware customer risk tiering
- explainability for global and local decisions
- actionable intervention simulation
- business-aware targeting support

Proposed Pipeline

The final fixed pipeline executed in the experiment was:

Stage 1: Data audit and cleaning

- loaded the raw Telco churn data
- checked data types, blanks, duplicates, and target distribution
- converted TotalCharges to numeric
- preserved the raw schema for later deployment

Stage 2: Train-test splitting

- 80% development set
- 20% final holdout test set
- stratified by target

Stage 3: Feature engineering

New features were created in five groups:

Monetary and tenure features

- avg_charge_per_month
- charge_tenure_ratio
- high_monthly_charge_flag
- low_tenure_flag
- long_tenure_flag

Service bundle features

- service_count
- support_service_count
- streaming_service_count
- internet_dependent_flag
- no_support_bundle_flag

Contract and payment risk features

- contract_risk_score
- payment_friction_score
- paperless_auto_payment_mismatch
- month_to_month_high_risk_flag

Interaction features

- tenure_x_monthlycharges
- internetservice_x_techsupport
- seniorcitizen_x_dependents
- contract_x_paymentmethod
- support_bundle_x_monthlycharges

Segment feature

- customer_segment_kmeans

Stage 4: Customer segmentation

A KMeans clustering step was added using numeric and engineered variables. The cluster label was included as a feature. The resulting segments showed meaningful churn separation:

- one cluster had very high churn risk
- one cluster had moderate churn risk
- two clusters had lower churn risk

Stage 5: Model training

Seven fixed models were used:

Baselines

- Logistic Regression
- Random Forest

Strong tabular ensemble models

- XGBoost

- LightGBM
- CatBoost

Deep tabular models

- TabNet
- FT-Transformer

Stage 6: Stacked ensemble

The final stack used:

- CatBoost
- XGBoost
- LightGBM
- TabNet
- FT-Transformer

A Logistic Regression meta-learner was trained on out-of-fold probabilities.

Stage 7: Calibration

The stacked ensemble was calibrated using:

- Platt scaling
- Isotonic regression

Isotonic regression achieved the best Brier score and was selected as the final calibrator.

Stage 8: Threshold optimization

The final calibrated model was evaluated across thresholds from 0.05 to 0.95. The best F1 threshold was found to be:

- **0.23**

Stage 9: Uncertainty-aware tiering

Customers were assigned to three groups:

- confident churn
- uncertain
- confident non-churn

Stage 10: Explainability

CatBoost was used for SHAP-based interpretation:

- global SHAP summary
- local SHAP waterfall explanations

Stage 11: Counterfactual simulation

Action-constrained interventions were tested for selected high-risk customers:

- contract to one year
- contract to two year
- add tech support
- add online security
- switch payment method
- reduce monthly charges by 10%
- bundle support and security

Stage 12: Business-aware targeting

A top-k targeting policy analysis was performed with a simple utility setup:

- intervention cost = 1.0
- retention benefit for correctly targeted churner = 5.0

Novelty of the Experiment

This work was intentionally designed to be more than a standard classifier comparison. The novelty comes from three main contributions.

Novelty 1: Calibrated and uncertainty-aware churn scoring

Most churn models output class labels or raw probabilities. In this work, the stacked ensemble was calibrated and then used for uncertainty-aware tiering. This produced operationally meaningful groups:

- confident churn
- uncertain
- confident non-churn

This makes the system more suitable for real decision support.

Novelty 2: Actionable explainability

Instead of stopping at feature importance, this work connected explanation to intervention. SHAP identified major churn drivers globally and locally, while intervention simulation tested how realistic changes affect churn probability.

Novelty 3: Business-aware retention targeting

Instead of treating classification as the final output, the experiment evaluated top-k targeting strategies using utility-based reasoning. This helps connect model output with operational retention decisions.

Exploratory Data Analysis Findings

Several strong patterns appeared before modeling.

Contract type

Month-to-month customers had far higher churn than longer-term contract customers.

Internet service

Fiber optic customers had the highest churn rate.

Payment method

Electronic check users showed the highest churn among payment methods.

Support and security services

Customers without:

- TechSupport
- OnlineSecurity

had much higher churn than customers with those services.

Numeric patterns

Compared with non-churners, churners had:

- much lower tenure
- higher monthly charges
- lower total charges

These findings were later confirmed again by SHAP.

Models Used and Their Interpretation

Logistic Regression

Result

- Accuracy: **0.7452**
- Precision: **0.5130**
- Recall: **0.7914**
- F1: **0.6225**
- ROC-AUC: **0.8463**
- PR-AUC: **0.6627**

Interpretation

This model performed surprisingly strongly, especially in ranking quality. It achieved the best PR-AUC among all individual models and one of the best ROC-AUC values. Because class_weight="balanced" was used, it emphasized recall, which made it suitable for identifying churners.

Meaning

A strong linear baseline remains highly competitive on this dataset.

Random Forest

Result

- Accuracy: **0.7736**
- Precision: **0.5696**
- Recall: **0.6016**
- F1: **0.5852**
- ROC-AUC: **0.8362**
- PR-AUC: **0.6420**

Interpretation

Random Forest achieved better accuracy than Logistic Regression but underperformed in recall, PR-AUC, and F1. It was more conservative and less effective at capturing churners.

Meaning

Random Forest is stable, but not one of the top-performing models in this experiment.

XGBoost

Result

- Accuracy: **0.7566**
- Precision: **0.5280**
- Recall: **0.7807**
- F1: **0.6300**
- ROC-AUC: **0.8412**
- PR-AUC: **0.6545**

Interpretation

XGBoost produced strong recall and a strong F1-score. It was one of the best overall classical nonlinear models and performed well as part of the final stack.

Meaning

XGBoost is a reliable core model for churn prediction.

LightGBM

Result

- Accuracy: **0.7637**
- Precision: **0.5408**
- Recall: **0.7273**
- F1: **0.6203**
- ROC-AUC: **0.8344**
- PR-AUC: **0.6338**

Interpretation

LightGBM performed adequately but did not outperform XGBoost or CatBoost. In the final stack, its coefficient contribution was also weak.

Meaning

Useful as a diversity component, but not dominant.

CatBoost

Result

- Accuracy: **0.7608**
- Precision: **0.5337**
- Recall: **0.7834**
- F1: **0.6349**
- ROC-AUC: **0.8455**
- PR-AUC: **0.6624**

Interpretation

CatBoost was the best single nonlinear model by F1-score. It also had very strong ROC-AUC and PR-AUC. Because it handles categorical patterns effectively, it became the most practical model for explainability.

Meaning

CatBoost is one of the strongest models in the entire experiment.

TabNet

Result

- Accuracy: **0.7317**
- Precision: **0.4967**
- Recall: **0.8102**
- F1: **0.6159**
- ROC-AUC: **0.8359**
- PR-AUC: **0.6505**

Interpretation

TabNet achieved the highest recall among all individual models. However, its precision and overall ranking performance were slightly below the best models.

Meaning

TabNet is useful when recall is especially important.

FT-Transformer

Result

- Accuracy: **0.7814**

- Precision: **0.7578**
- Recall: **0.2594**
- F1: **0.3865**
- ROC-AUC: **0.8469**
- PR-AUC: **0.6617**

Interpretation

FT-Transformer had the highest ROC-AUC among individual models and very high precision at the default threshold. However, its recall was very low, meaning it was very conservative at threshold 0.5.

Meaning

The model ranked customers very well, but needed threshold adjustment or stacking to become operationally useful.

Stacked Ensemble Results

Raw stacked ensemble

- Accuracy: **0.7573**
- Precision: **0.5289**
- Recall: **0.7834**
- F1: **0.6315**
- ROC-AUC: **0.8463**
- PR-AUC: **0.6619**

Meta-learner coefficients

- CatBoost: strongest positive contribution
- XGBoost: second strongest
- FT-Transformer: meaningful contribution
- TabNet: smaller positive contribution
- LightGBM: minimal / weak contribution

Interpretation

The stack was competitive but did not clearly dominate the best individual models before calibration. However, its main value appeared later after calibration and threshold optimization.

Calibration Results

Three versions were compared:

- Uncalibrated
- Platt
- Isotonic

Calibration comparison

- Best Brier score: **Isotonic**
- Platt was extremely close
- Uncalibrated had slightly stronger ROC-AUC / PR-AUC, but worse probability quality

Selected calibrator

Isotonic regression

Interpretation

Calibration substantially improved the reliability of probabilities, which was necessary for:

- threshold optimization
- uncertainty tiering
- business targeting

Final Operational Model

The final operational model was:

Calibrated stacked ensemble with threshold = 0.23

Final results

- Threshold: **0.23**
- Accuracy: **0.7438**
- Precision: **0.5106**
- Recall: **0.8342**
- F1: **0.6335**
- ROC-AUC: **0.8458**
- PR-AUC: **0.6568**
- Brier score: **0.1348**

Interpretation

This threshold produced a very good operational balance for churn retention:

- high recall
- strong F1
- calibrated probability quality preserved

This final setting outperformed the calibrated default threshold of 0.5 for churn-capture purposes.

Uncertainty-Aware Results

Customers were partitioned as follows:

Tier counts

- uncertain: **819**
- confident_non_churn: **483**
- confident_churn: **107**

Tier quality

confident_churn

- count: **107**
- actual churn rate: **0.8131**
- average predicted probability: **0.8332**

uncertain

- count: **819**
- actual churn rate: **0.3309**

confident_non_churn

- count: **483**
- actual churn rate: **0.0331**
- correct non-churn rate: **0.9669**

Interpretation

This is one of the strongest parts of the experiment. The uncertainty layer successfully separated:

- high-confidence churners

- high-confidence non-churners
- ambiguous middle cases

This is useful for operational triage.

Business-Aware Targeting Results

Using the assumed utility setup:

- intervention cost = 1.0
- benefit from correctly targeted churner = 5.0

Best net-utility policy

- top-k percent: **50%**
- customers targeted: **704**
- true positives: **326**
- false positives: **378**
- recall at k: **0.8717**
- precision at k: **0.4631**
- net utility: **926.0**

Interpretation

Under the current assumptions, broader targeting remained profitable because the benefit-to-cost ratio was generous. However:

- top 5% gave the best precision
- top 20–25% gave a balanced operational policy
- top 50% maximized modeled utility

This demonstrates how policy choice depends on business assumptions rather than only classifier score.

Explainability Results

Global SHAP Findings

Top global features by mean absolute SHAP value:

1. Contract
2. InternetService
3. charge_tenure_ratio
4. OnlineSecurity
5. contract_risk_score
6. TechSupport
7. tenure
8. PaymentMethod
9. MultipleLines
10. PaperlessBilling

Interpretation

The SHAP results are highly consistent with both EDA and telecom domain intuition.

Important insights:

- contract structure is central to churn
- service support and security matter strongly
- price relative to tenure is highly informative
- engineered features contributed significantly, especially:
    - charge_tenure_ratio
    - contract_risk_score
    - month_to_month_high_risk_flag
    - internetservice_x_techsupport

This validates the engineered feature design.

Local SHAP Findings

Selected top-risk customers had highly consistent local drivers:

- high charge_tenure_ratio
- InternetService = Fiber optic
- Contract = Month-to-month

Interpretation

The local explanations show that many very high-risk customers were:

- new customers
- on expensive plans
- using fiber optic service
- on month-to-month contracts
- often lacking security/support features

Counterfactual Intervention Results

Five very high-risk test customers were selected. For each one, realistic interventions were simulated.

Best intervention for all five cases

contract_to_two_year

Example probability reductions

- 0.5763
- 0.3974
- 0.4109
- 0.3328
- 0.4406

Other useful interventions

- bundle_support_security
- contract_to_one_year

Weak interventions

- payment method switch alone
- small monthly charge reduction alone

Interpretation

The strongest counterfactual recommendation across all selected cases was moving to a longer-term contract, especially a two-year contract. This aligns with:

- EDA
- SHAP

- business intuition

This is a strong actionable result because it connects:

- model output
- explanation
- intervention simulation
- measurable churn risk reduction

Why the Final Pipeline Is Strong

This pipeline stands out because it does not treat churn prediction as a single-step classifier problem. Instead, it combines:

- feature-rich tabular modeling
- strong ensemble and deep-tabular baselines
- calibrated probabilities
- threshold optimization
- uncertainty-aware customer segmentation
- explainable AI
- intervention simulation
- business-aware targeting

That makes it not only predictive, but operational.

Strengths of the Experiment

Methodological strengths

- multiple model families compared
- final holdout test set used
- calibration explicitly evaluated
- threshold optimization conducted
- stack built from out-of-fold predictions
- uncertainty and intervention layers included

Practical strengths

- usable for retention decision support
- explainable to business stakeholders
- supports policy simulation
- suitable for web-app deployment

Research strengths

- novelty comes from combining:
    - calibration
    - uncertainty
    - counterfactual recourse
    - business targeting

    rather than only comparing classifiers

Limitations

Despite the strong results, there are limitations.

Dataset limitations

- relatively small dataset
- static tabular data only
- no temporal sequence of behavior
- no real intervention outcome labels

Business limitations

- business utility assumptions were simulated, not sourced from real telecom cost data
- counterfactual actions were rule-based simulations, not causal proof of retention success

Modeling limitations

- nested cross-validation with full tuning for every model was not fully executed in the final deployed notebook path
- the final stack did not dominate every metric, though it became most useful after calibration and threshold tuning
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Final Results Table

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.7452	0.5130	0.7914	0.6225	0.8463	0.6627
Random Forest	0.7736	0.5696	0.6016	0.5852	0.8362	0.6420
XGBoost	0.7566	0.5280	0.7807	0.6300	0.8412	0.6545
LightGBM	0.7637	0.5408	0.7273	0.6203	0.8344	0.6338
CatBoost	0.7608	0.5337	0.7834	0.6349	0.8455	0.6624
TabNet	0.7317	0.4967	0.8102	0.6159	0.8359	0.6505
FT-Transformer	0.7814	0.7578	0.2594	0.3865	0.8469	0.6617
Stacked Ensemble	0.7573	0.5289	0.7834	0.6315	0.8463	0.6619
Calibrated Stacked Ensemble (0.50)	0.8077	0.6746	0.5321	0.5949	0.8458	0.6568
Final Calibrated Stacked Ensemble (0.23)	0.7438	0.5106	0.8342	0.6335	0.8458	0.6568